

Effect of Baseline ASPECTS on Tenecteplase Efficacy Before Thrombectomy in Acute Large-Vessel Occlusion Stroke

A Post Hoc Analysis of the BRIDGE-TNK Randomized Trial

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Abstract

Background and Objectives

The impact of ischemic extent on the efficacy and safety of intravenous thrombolysis before thrombectomy remains uncertain. The aim of this study was to evaluate whether the baseline ischemic extent, measured by the Alberta Stroke Program Early Computed Tomography Score (ASPECTS), modifies outcomes of intravenous tenecteplase administered before endovascular thrombectomy.

Methods

This was a post hoc analysis of the BRIDGE-TNK (thrombectomy with vs without rhTNK-tPA in stroke) trial, conducted across China from May 2022 to September 2024. We compared the efficacy and safety of intravenous tenecteplase plus thrombectomy vs thrombectomy alone in acute large-vessel occlusion stroke patients within 4.5 hours of last known well, stratified by baseline ASPECTS (<8 vs 8–10). The outcomes included 90-day functional independence (modified Rankin Scale score of 0–2), 48-hour symptomatic intracranial hemorrhage (sICH), and 90-day mortality. Regression models incorporating a treatment-by-ASPECTS interaction term were used for analysis.

Results

Among 550 patients, 241 (43.8%) had ASPECTS <8 (median [interquartile range, IQR] age, 69 [61–77] years; 56.4% male) and 309 had ASPECTS 8–10 (median [IQR] age, 70 [61–77] years; 59.5% male). The rate of functional independence was significantly higher in the tenecteplase plus thrombectomy group than in the thrombectomy-alone group in the ASPECTS <8 subgroup (adjusted risk ratio [aRR], 1.67; 95% CI 1.18–2.35), but not in the ASPECTS 8–10 subgroup (aRR, 0.99; 95% CI 0.84–1.17; $p_{\text{interaction}} = 0.007$). Rates of sICH did not differ significantly between treatment groups in either ASPECTS subgroups (ASPECTS <8: 10.0% vs 11.2%; ASPECTS 8–10: 7.5% vs 2.8%; $p_{\text{interaction}} = 0.11$). Ninety-day mortality was comparable between treatment groups in the ASPECTS <8 subgroup, but numerically higher with tenecteplase plus thrombectomy in the ASPECTS 8–10 subgroup (aRR = 1.89, 95% CI 0.99–3.61, $p_{\text{interaction}} = 0.04$).

Discussion

In this exploratory post hoc analysis, a signal of benefit was observed in patients with ASPECTS <8 who received intravenous tenecteplase before thrombectomy, whereas no

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Glossary

aRR = adjusted risk ratio; **ASPECTS** = Alberta Stroke Program Early Computed Tomography Score; **EVT** = endovascular thrombectomy; **IQR** = interquartile range; **IVT** = intravenous thrombolysis; **MCA** = middle cerebral artery; **mRS** = modified Rankin Scale; **NCCT** = noncontrast CT; **NIHSS** = National Institutes of Health Stroke Scale; **PCA** = posterior cerebral artery; **RCT** = randomized controlled trial; **sICH** = symptomatic intracranial hemorrhage.

functional improvement and possible safety concerns were seen in those with ASPECTS 8–10. Prospective confirmation in randomized trials is required before practice change.

Trial Registration Information

ClinicalTrials.gov; Unique identifier: NCT04733742.

Introduction

Six randomized controlled trials (RCTs) suggest that intravenous thrombolysis (IVT) before endovascular thrombectomy (EVT), compared with thrombectomy alone, improves outcomes modestly in a time-dependent manner, with statistical evidence of benefit no longer present if initiated beyond 140 minutes from last known well.^{1–7} In these trials, alteplase was used almost exclusively as the thrombolytic agent, yet it exhibited a low rate of prethrombectomy reperfusion of approximately 4%.⁸ Tenecteplase, a genetically modified variant of alteplase with increased fibrin specificity, exhibits a more potent thrombolytic effect and can be administered by rapid bolus injection.⁹ This approach minimizes treatment delays⁸ and has been suggested in several studies as potentially improving recanalization rates,^{10–13} thereby possibly enhancing the overall effectiveness of bridging therapy.

The extent of ischemic region before treatment is a critical factor influencing the administration of IVT before EVT.^{14,15} Patients presenting with unfavorable Alberta Stroke Program Early Computed Tomography Score (ASPECTS) are often considered for direct thrombectomy alone because of concerns regarding increased hemorrhagic risks with thrombolytic agents. Although the aforementioned 6 RCTs did not apply strict imaging inclusion criteria, the enrolled populations predominantly had favorable baseline ASPECTS (median 9 with interquartile range [IQR] 7 to 10),⁸ leaving the benefit-risk profile of thrombolysis in patients with unfavorable ASPECTS uncertain.

Recently, we reported the BRIDGE-TNK (thrombectomy with vs without recombinant human tenecteplase) trial showing the benefit of intravenous tenecteplase before thrombectomy for patients with acute ischemic stroke due to large-vessel occlusion who had presented within 4.5 hours of onset.¹⁶ We thus compared outcomes between the 2 treatment strategies (tenecteplase plus thrombectomy vs thrombectomy alone) in patients with ASPECTS <8 and those with ASPECTS 8–10 to evaluate whether baseline ischemic extent

modifies the efficacy and safety of administering tenecteplase before EVT.

Methods

Participants

This study was a post hoc subgroup analysis of the BRIDGE-TNK trial, which was a multicenter, investigator-initiated, open-label randomized study designed to evaluate the efficacy and safety of intravenous tenecteplase before thrombectomy vs thrombectomy alone across 39 thrombectomy-capable centers throughout China.¹⁶ Patients with acute large-vessel occlusion stroke eligible for IVT within 4.5 hours of last-known-well time and scheduled for EVT were enrolled. The detailed inclusion and exclusion criteria have been described previously.¹⁷

Standard Protocol Approvals, Registrations, and Patient Consents

The study adhered to the principles of the Declaration of Helsinki. The protocol received approval from the ethics committees of the Second Affiliated Hospital of Army Medical University (Xinqiao Hospital) and all participating centers. Written informed consent was obtained from all patients or their legal representatives before randomization. The study is registered at ClinicalTrials.gov as BRIDGE-TNK, NCT04733742.

Clinical Assessment

Baseline characteristics, including demographics, medical history, blood pressure, prestroke mRS score, baseline National Institutes of Health Stroke Scale (NIHSS) score, occlusion site, laboratory tests, type of stroke, and time metrics were provided by the site investigators and prospectively collected. An independent imaging core laboratory blinded to clinical characteristics and outcomes assessed the occlusion site, collateral status (American Society of Interventional and Therapeutic Neuroradiology/Society of Interventional Radiology, ASITN/SIR),¹⁸ reperfusion level by expanded treatment in cerebral ischemia (eTICI),^{19,20} and intracranial hemorrhage.²¹

The ASPECTS was centrally assessed using the F-stroke software (Version 1.0.42, Neuroblem Ltd. Company, Shanghai, China) at our imaging core laboratory on baseline non-contrast CT (NCCT) or diffusion-weighted MRI (DWI). All ASPECTS assessments were reviewed by a senior neurologist, with any discrepancies resolved by the neurologist's adjudication. For anterior circulation, the ASPECTS for the middle cerebral artery (MCA) territory was used, which allots a base score of 10, with 1 point subtracted for each predefined MCA region that shows early ischemic change.²² For posterior circulation, the pc-ASPECTS was used, which assigns a base score of 10, with 1 point subtracted for early ischemic change in each of the left or right thalamus, cerebellum, or posterior cerebral artery (PCA) territory and 2 points subtracted for early ischemic change in any part of the midbrain or pons.²³ A score of 10 indicates no visible early ischemic change, whereas lower scores reflect progressively more extensive ischemic involvement. In this subgroup analysis, ASPECTS was stratified by < 8 vs 8–10. This cutoff was chosen in a post hoc, data-driven manner (not prespecified) to better distinguish clinical distinction between favorable and unfavorable ischemic profiles.^{24,25}

Outcomes

Consistent with the main trial report,¹⁶ the primary outcome in this study was the proportion of patients achieving functional independence (defined as a modified Rankin Scale [mRS] score of 0–2) at 90 days. Secondary outcomes included level of disability (mRS scores) at 90 days, the proportion of patients achieving freedom from disability (mRS 0 to 1) at 90 days, and the proportion of patients achieving ambulation and independent self-care (mRS 0–3) at 90 days. Radiologic outcomes were successful reperfusion, defined as an eTICI grade of 2b50, 2c, or 3 (ranging from 0, indicating no reperfusion, to 3, indicating complete reperfusion) on angiography performed before and after EVT, and first-pass reperfusion (defined as an eTICI grade of 2c or 3 after the initial thrombectomy pass).^{19,20}

Safety outcomes comprised symptomatic intracranial hemorrhage (sICH) classified according to the modified Heidelberg Bleeding Classification within 48 hours,²¹ any ICH detected on imaging within 48 hours,²¹ and mortality within 90 days. Procedure-related complications and serious adverse events were also compared.

Statistical Analysis

Continuous variables were expressed as either means with standard deviations or medians with IQRs, depending on their distribution, and compared using the Student *t* test or Wilcoxon rank-sum test. Categorical variables were presented as numbers and percentages and compared using the χ^2 test or Fisher exact test. There were no missing data for the key baseline characteristics and the primary outcome, and no imputation was performed.

Binary outcomes between treatment groups were evaluated using modified Poisson regression, adjusted for 5 variables: age, baseline NIHSS score, extent of ischemic injury (ASPECTS), occlusion site, and time from last known well to randomization. Adjusted risk ratios (aRRs) with 95% CIs were reported. A proportional odds logistic regression was used to estimate the common odds ratio with 95% CI for a favorable shift in the 90-day mRS distribution. The Cox proportional hazards regression was used to calculate the hazard ratio with 95% CI for 90-day mortality. Effect modifications by ASPECTS strata were examined by including a treatment-by-ASPECTS interaction term into the regression models.

Propensity score-matching methods were conducted to balance important covariates. The propensity score was estimated using a multivariable logistic regression model, with the treatment received as the dependent variable and age, sex, medical history, prestroke mRS, systolic blood pressure, glucose, baseline NIHSS score, baseline ASPECTS, stroke etiology, occlusion sites, hemisphere, collateral score, and onset-to-randomization time as covariates. A 1:1 nearest-neighbor matching algorithm with a caliper of 0.2 was applied. The standardized mean differences were calculated, and a threshold of 0.15 or below was regarded as indicative of acceptable balance. Additional sensitivity analyses were performed in patients with extensive ischemic core (ASPECTS 0–5) and in those with isolated anterior circulation large-vessel occlusion using only NCCT-based ASPECTS.

All statistical analyses were performed using SPSS (IBM Statistics Version 26.0) and R software (R Foundation for Statistical Computing, version 4.1.3). All reported *p* values are two-sided, and *p* values less than 0.05 are considered statistically significant.

Data Availability

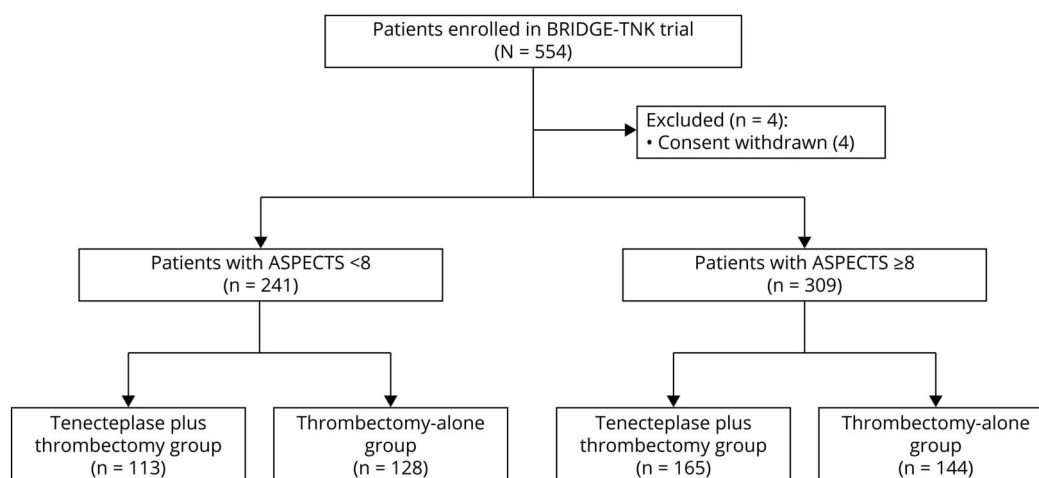
Data from the BRIDGE-TNK study will be made available at the request of qualified investigators whose proposal of data use has been approved by an independent review committee.

Results

Patient Characteristics

Enrollment took place from May 2022 to September 2024, with follow-up completed in December 2024. Among the 554 enrolled patients, 4 patients withdrew consent, leaving 550 patients in this analysis (Figure 1). The inter-rater reliability for ASPECTS scoring was good, with an intraclass correlation coefficient (ICC) of 0.834. The frequency distribution of ASPECTS is presented in eFigure 1. Of these, 241 (43.8%) had an ASPECTS <8 (median [IQR] age, 69 [61–77] years; 136 male individuals [56.4%]) and 309 had an ASPECTS of 8–10 (median [IQR] age, 70 [61–77] years; 184 male individuals [59.5%]).

Figure 1 Flowchart of Patient Selection



ASPECTS = Alberta Stroke Program Early Computed Tomography Score.

Patients with an ASPECTS <8 had higher baseline NIHSS scores (median 18 vs 15; $p = 0.01$), higher blood glucose levels (median 7.6 vs 7.1; $p = 0.01$), higher proportion of internal carotid artery occlusion (38.6% vs 23.0%; $p < 0.001$), higher proportion of MRI assessment (15.8% vs 8.7%; $p = 0.01$), longer time from onset to randomization (median 172 vs 153 minutes; $p = 0.01$), and worse collateral scores (ASTIN/SIR 0–2: 80.8% vs 71.2%; $p = 0.01$), compared with patients with an ASPECTS 8–10 (eTable 1). The baseline characteristics and covariates between the treatment arms stratified by ASPECTS are listed in Table 1. The proportions of rescue therapy and IV tirofiban after thrombectomy were numerically higher in patients treated with thrombectomy alone than those receiving tenecteplase plus thrombectomy, but the differences were statistically significant only in the ASPECTS <8 subgroup.

Outcomes

Patients with an ASPECTS ≥8 attained significantly better outcomes relative to those with an ASPECTS <8, including higher rates of favorable mRS outcomes (0–2, 0–1, and 0–3), a shift toward lower mRS scores, as well as lower all-cause mortality, sICH, and any ICH (eTable 2). In the ASPECTS <8 subgroup, 47 (41.6%) patients in the tenecteplase plus thrombectomy group and 32 (25.0%) in the thrombectomy-alone group achieved a mRS score of 0–2 (adjusted risk ratio [aRR], 1.67; 95% CI 1.18–2.35; $p = 0.003$; Table 2). By contrast, no significant difference was observed in the ASPECTS 8–10 subgroup (60.6% vs 61.1%; aRR, 0.99; 95% CI 0.84–1.17; $p = 0.92$), with significant treatment-by-ASPECTS interaction ($p_{\text{interaction}} = 0.007$). Tenecteplase plus thrombectomy was also associated with a favorable shift in the overall mRS distribution only in patients with ASPECTS <8, but not in patients with ASPECTS 8–10 ($p_{\text{interaction}} = 0.01$; Figure 2 and Table 2). Significant effect modification by ASPECTS stratification

was noted for the mRS of 0–1 ($p_{\text{interaction}} = 0.01$) and mRS of 0–3 ($p_{\text{interaction}} = 0.04$).

There were numerically higher rates of successful reperfusion before thrombectomy with tenecteplase plus thrombectomy vs thrombectomy alone in both ASPECTS subgroups (<8: 6.2% vs 1.6%; ≥8: 6.1% vs 0.7%; $p_{\text{interaction}} = 0.52$). No significant interaction was observed for other radiologic outcomes.

sICH occurred in 11 (10.0%) patients in the tenecteplase plus thrombectomy group vs 14 (11.2%) patients in the thrombectomy-alone group in the subgroup of ASPECTS <8 (aRR, 0.92; 95% CI 0.42–1.99). In the ASPECTS 8–10 subgroup, the number of sICH was 12 (7.5%) vs 4 (2.8%), respectively (aRR, 2.87; 95% CI 0.94–8.78; $p_{\text{interaction}} = 0.11$; Table 2). Tenecteplase before thrombectomy was not associated with an increased risk of any ICH in both ASPECTS subgroups ($p_{\text{interaction}} = 0.53$). Mortality at 90 days was similar between 2 treatment arms in the ASPECTS <8 subgroup (29.2% vs 30.5%) but was numerically higher in the tenecteplase plus thrombectomy group in the ASPECTS 8–10 subpopulation (17.6% vs 10.4%; aRR, 1.89; 95% CI 0.99–3.61; $p_{\text{interaction}} = 0.04$; Table 2). Serious adverse events and procedure-related complications are presented in eTable 3.

Figure 3 displayed the association between treatment modality and ASPECTS as a continuous variable. There was a significant interaction effect between treatment and ASPECTS score for 90-day functional independence ($p_{\text{interaction}} = 0.03$), with a crossover at ASPECTS 9. No significant interaction was observed for the risk of sICH ($p_{\text{interaction}} = 0.27$), despite a crossover was seen at ASPECTS 5.

Clinical outcomes were also compared between treatment groups in patients with ASPECTS 0–5, as presented in

Table 1 Patient Characteristics and Covariates by Treatment and ASPECTS Stratification

Variables, no. (%)	ASPECTS <8 N = 241			ASPECTS ≥8 N = 309		
	Tenecteplase plus thrombectomy (n = 113)	Thrombectomy alone (n = 128)	p Value	Tenecteplase plus thrombectomy (n = 165)	Thrombectomy alone (N = 144)	p Value
Age, median (IQR), y	70 (61–79)	69 (62–75)	0.33	70 (60–77)	70 (63–77)	0.37
Sex, male	63 (55.8)	73 (57.0)	0.84	98 (59.4)	86 (59.7)	0.95
Medical history						
Hypertension	68 (60.2)	92 (71.9)	0.06	103 (62.4)	86 (59.7)	0.63
Diabetes	27 (23.9)	31 (24.2)	0.95	30 (18.2)	27 (18.8)	0.90
Hyperlipidemia	20 (17.7)	37 (28.9)	0.04	33 (20.0)	30 (20.8)	0.86
Atrial fibrillation	51 (45.1)	44 (34.4)	0.06	50 (30.3)	52 (36.1)	0.28
Ischemic stroke	25 (22.1)	25 (19.5)	0.62	27 (16.4)	25 (17.4)	0.88
Coronary heart disease	23 (20.4)	35 (27.3)	0.21	35 (21.2)	38 (26.4)	0.29
Prestroke mRS >1	0	4 (3.1)	0.13	4 (2.4)	1 (0.7)	0.38
Baseline NIHSS score, median (IQR)	18 (12–22)	17 (12–20)	0.32	15 (12–20)	15 (11–20)	0.75
Baseline SBP, median (IQR), mm Hg	144 (130–160)	150 (132–170)	0.22	143 (130–166)	148 (126–168)	0.79
Baseline glucose, median (IQR), mmol/L ^a	7.5 (6.1–9.7)	7.6 (6.4–9.4)	0.82	7.1 (6.0–8.8)	7.0 (5.9–8.5)	0.50
ASPECTS assessment mode			0.95			0.52
CT	95 (84.1)	108 (84.4)		149 (90.3)	133 (92.4)	
MR	18 (15.9)	20 (15.6)		16 (9.7)	11 (7.6)	
Ischemic location			0.56			0.12
Left hemisphere	44 (38.9)	55 (43.0)		67 (40.6)	75 (52.1)	
Right hemisphere	58 (51.3)	65 (50.8)		79 (47.9)	57 (39.6)	
Posterior circulation	11 (9.7)	8 (6.3)		19 (11.5)	12 (8.3)	
Occlusion site			0.50			0.45
Internal carotid artery	40 (35.4)	53 (41.4)		37 (22.4)	34 (23.6)	
Middle cerebral artery M1	56 (49.6)	63 (49.2)		98 (59.4)	82 (56.9)	
Middle cerebral artery M2	7 (6.2)	4 (3.1)		11 (6.7)	16 (11.1)	
Vertebrobasilar artery	10 (8.8)	8 (6.3)		19 (11.5)	12 (8.3)	
Stroke classification			0.54			0.74
Atherothrombotic	29 (25.7)	40 (31.3)		53 (32.1)	45 (31.3)	
Cardioembolic	70 (61.9)	76 (59.4)		102 (61.8)	87 (60.4)	
Undetermined or others	14 (12.4)	12 (9.4)		10 (6.1)	12 (8.3)	
Duration time, median (IQR) min						
From onset to hospital	107 (63–152)	115 (75–148)	0.68	79 (45–121)	80 (45–129)	0.66
From onset to randomization	174 (124–219)	171 (136–221)	0.50	149 (118–213)	162 (124–211)	0.38

Continued

Table 1 Patient Characteristics and Covariates by Treatment and ASPECTS Stratification (*continued*)

Variables, no. (%)	ASPECTS <8 N = 241			ASPECTS ≥8 N = 309		
	Tenecteplase plus thrombectomy (n = 113)	Thrombectomy alone (n = 128)	p Value	Tenecteplase plus thrombectomy (n = 165)	Thrombectomy alone (N = 144)	p Value
From door to groin puncture ^b	83 (63–111)	84 (66–107)	0.70	91 (72–118)	85 (65–113)	0.17
From groin puncture to reperfusion ^b	55 (34–83)	65 (42–118)	0.02	56 (36–87)	62 (40–95)	0.27
Collateral score (ASTIN/ SIR) ^b			0.44			0.10
Poor 0–2	89 (78.8)	105 (82.7)		124 (75.2)	96 (66.7)	
Good 3–4	24 (21.2)	22 (17.3)		41 (24.8)	48 (33.3)	
Thrombectomy attempts ^c	2 (1–2)	2 (1–3)	0.18	1 (1–2)	1 (1–2)	0.65
Rescue therapy ^d	15 (14.3)	31 (25.4)	0.04	36 (23.1)	41 (29.3)	0.22
IV tirofiban after thrombectomy	21 (18.6)	45 (35.2)	0.004	47 (28.5)	50 (34.7)	0.24

Abbreviations: ASPECTS = Alberta Stroke Program Early Computed Tomography Score; ASITN/SIR = American Society of Interventional and Therapeutic Neuroradiology/Society of Interventional Radiology; IQR = interquartile range; MR = magnetic resonance; mRS = modified Rankin scale; NA = not applicable; NIHSS = National Institutes of Health Stroke Scale; SBP = systolic blood pressure.

^a For the glucose level, data were missing for 7 patients (2 in the ASPECTS <8 group and 5 in the ASPECTS ≥8 group). To convert the values to milligrams per deciliter, divide by 0.05551.

^b Data were not available for 1 patient in the ASPECTS <8 group.

^c Data were not available for 26 patients (14 in the ASPECTS <8 group and 12 in the ASPECTS ≥8 group).

^d Data were not available for 27 patients (14 in the ASPECTS <8 group and 13 in the ASPECTS ≥8 group).

eTable 4. The tenecteplase plus thrombectomy group showed a numerically higher rate of functional independence (27.3% vs 11.1%; $p = 0.07$) and a significantly higher proportion of patients with mRS of 0–1 (18.2 vs 2.2%; $p = 0.04$), alongside a favorable shift in overall mRS distribution (median [IQR], 4 [2–6] vs 6 [3–6]; $p = 0.02$), compared with the thrombectomy-alone group. Rates of sICH were comparable between the 2 groups (15.6% vs 14.3%; $p = 1.00$). Notably, none of the 12 participants with an extremely low ASPECTS (0–2) achieved functional independence at 90 days.

Sensitivity Analysis

To test the robustness of the treatment-by-ASPECTS interaction, we performed another sensitivity analysis restricted to anterior circulation large-vessel occlusions using only NCCT-based ASPECTS. The analysis included 186 patients in the ASPECTS <8 subgroup and 253 patients in the ASPECTS 8–10 subgroup. In the ASPECTS <8 subgroup, the tenecteplase plus thrombectomy group showed a higher rate of functional independence (40.7% vs 23.0%; aRR, 1.70; 95% CI 1.11–2.59; $p = 0.01$). In the ASPECTS 8–10 subgroup, no significant difference was observed (62.6% vs 62.3%; aRR = 1.01; 95%CI 0.84–1.21; $p = 0.92$), with a significant interaction by ASPECTS ($p_{\text{interaction}} = 0.02$). In the ASPECTS 8–10 subgroup, the tenecteplase plus thrombectomy group exhibited significantly higher rates of prethrombectomy successful reperfusion (6.9% vs 0.8%), sICH (8.5% vs 2.5%), and 90-day mortality (17.6% vs 8.2%), whereas the rates of prethrombectomy reperfusion,

sICH, and mortality were comparable across treatment arms in the ASPECTS <8 subgroup (eTable 5).

Propensity Score Matching

After 1:1 propensity score matching (Table 3), 180 patients were included in the ASPECTS <8 subgroup and 220 patients in the ASPECTS 8–10 subgroup. Demographic characteristics, cardiovascular risk factors, stroke severity, time metrics, and imaging features were well matched between the 2 treatment arms in both ASPECTS subgroups.

In the ASPECTS <8 subgroup, intravenous tirofiban was administered after thrombectomy in 18 patients (20.0%) in the bridging therapy group compared with 30 patients (33.3%) in the thrombectomy-alone group ($p = 0.04$). The tenecteplase plus thrombectomy group showed a higher rate of functional independence (37.8% vs 20.0%; $p = 0.009$) and mRS 0–1 (22.2% vs 8.9%; $p = 0.01$). The sICH and mortality were similar between the 2 groups. In the ASPECTS 8–10 subgroup, 70 (63.6%) patients in the tenecteplase plus thrombectomy group achieved functional independence at 90 days, vs 66 (60.0%) patients in the thrombectomy-alone group ($p = 0.58$). The successful reperfusion before thrombectomy was significantly more frequent in the tenecteplase plus thrombectomy group (6.4% vs 0.9%, $p = 0.03$), but rates of sICH and mortality were numerically higher in this group (8.4% vs 2.7% and 17.3% vs 9.1%, respectively).

Table 2 Efficacy and Safety Outcomes by Treatment and ASPECTS Stratification

Outcome	Tenecteplase plus thrombectomy	Thrombectomy alone	p Value	Effect measures ^a	Adjusted treatment effect (95% CI) ^b	p Value	P _{interaction}
ASPECTS <8, no.	n = 113	n = 128					
Primary efficacy outcome, no. (%)							
mRS score of 0–2 at 90 d	47 (41.6)	32 (25.0)	0.006	Risk ratio	1.67 (1.18–2.35)	0.003	0.007
Secondary efficacy outcomes, no. (%)							
mRS score at 90 d, median (IQR)	3 (2–6)	4 (3–6)	0.05	Common odds ratio	1.82 (1.14–2.93)	0.01	0.01
mRS score of 0–1 at 90 d	28 (24.8)	15 (11.7)	0.01	Risk ratio	2.18 (1.27–3.73)	0.01	0.01
mRS score of 0–3 at 90 d	64 (56.6)	60 (46.9)	0.13	Risk ratio	1.21 (0.96–1.52)	0.10	0.04
Successful reperfusion before thrombectomy ^c	7 (6.2)	2 (1.6)	0.06	Risk ratio	3.67 (0.72–18.7)	0.12	0.52
Successful reperfusion after thrombectomy	101 (89.4)	119 (93.7)	0.23	Risk ratio	0.96 (0.88–1.04)	0.39	0.51
First-pass reperfusion ^d	39 (37.1)	40 (32.8)	0.49	Risk ratio	1.10 (0.77–1.57)	0.59	0.46
Safety outcomes, no. (%)							
Symptomatic ICH within 48 hours ^e	11 (10.0)	14 (11.2)	0.77	Risk ratio	0.92 (0.42–1.99)	0.82	0.11
Any evidence of ICH within 48 hours ^e	48 (43.6)	58 (46.4)	0.67	Risk ratio	0.99 (0.76–1.30)	0.97	0.53
Death within 90 d	33 (29.2)	39 (30.5)	0.83	Hazard ratio ^f	0.86 (0.54–1.38)	0.53	0.04
ASPECTS ≥8, no.	n = 165	n = 144					
Primary efficacy outcome, no. (%)							
mRS score of 0–2 at 90 d	100 (60.6)	88 (61.1)	0.93	Risk ratio	0.99 (0.84–1.17)	0.92	NA
Secondary efficacy outcomes, no. (%)							
mRS score at 90 d, median (IQR)	2 (1–4)	2 (0–4)	0.48	Generalized odds ratio	0.84 (0.56–1.25)	0.38	NA
mRS score of 0–1 at 90 d	69 (41.8)	61 (42.4)	0.92	Risk ratio	1.00 (0.78–1.28)	0.99	NA
mRS score of 0–3 at 90 d	114 (69.1)	108 (75.0)	0.25	Risk ratio	0.92 (0.81–1.06)	0.23	NA
Successful reperfusion before thrombectomy	10 (6.1)	1 (0.7)	0.01	Risk ratio	8.70 (1.10–68.75)	0.04	NA
Successful reperfusion after thrombectomy	153 (92.7)	136 (94.4)	0.54	Risk ratio	0.99 (0.93–1.05)	0.64	NA
First-pass reperfusion ^d	68 (43.6)	64 (45.4)	0.76	Risk ratio	0.96 (0.74–1.24)	0.74	NA
Safety outcomes, no. (%)							
Symptomatic ICH within 48 h ^e	12 (7.5)	4 (2.8)	0.07	Risk ratio	2.87 (0.94–8.78)	0.07	NA
Any evidence of ICH within 48 h ^e	36 (22.4)	29 (20.1)	0.64	Risk ratio	1.19 (0.77–1.84)	0.43	NA
Death within 90 d	29 (17.6)	15 (10.4)	0.07	Hazard ratio ^f	1.89 (0.99–3.61)	0.05	NA

Abbreviations: ASPECTS = Alberta Stroke Program Early Computed Tomography Score; ICH = intracranial hemorrhage; IQR = interquartile range; mRS = modified Rankin Scale; NA = not applicable; NIHSS = National Institutes of Health Stroke Scale.

^a Treatment effects are reported as relative risk (95% CI) for primary outcomes from modified Poisson regression, common odds ratio (95% CI) for the ordinal shift analysis toward a better outcome, and hazard ratio (95% CI) for death by a Cox proportional hazards regression model.

^b The treatment effect was adjusted for age, NIHSS score at baseline, ASPECTS at baseline, site of occlusion, and time from the last-known-well time point to randomization.

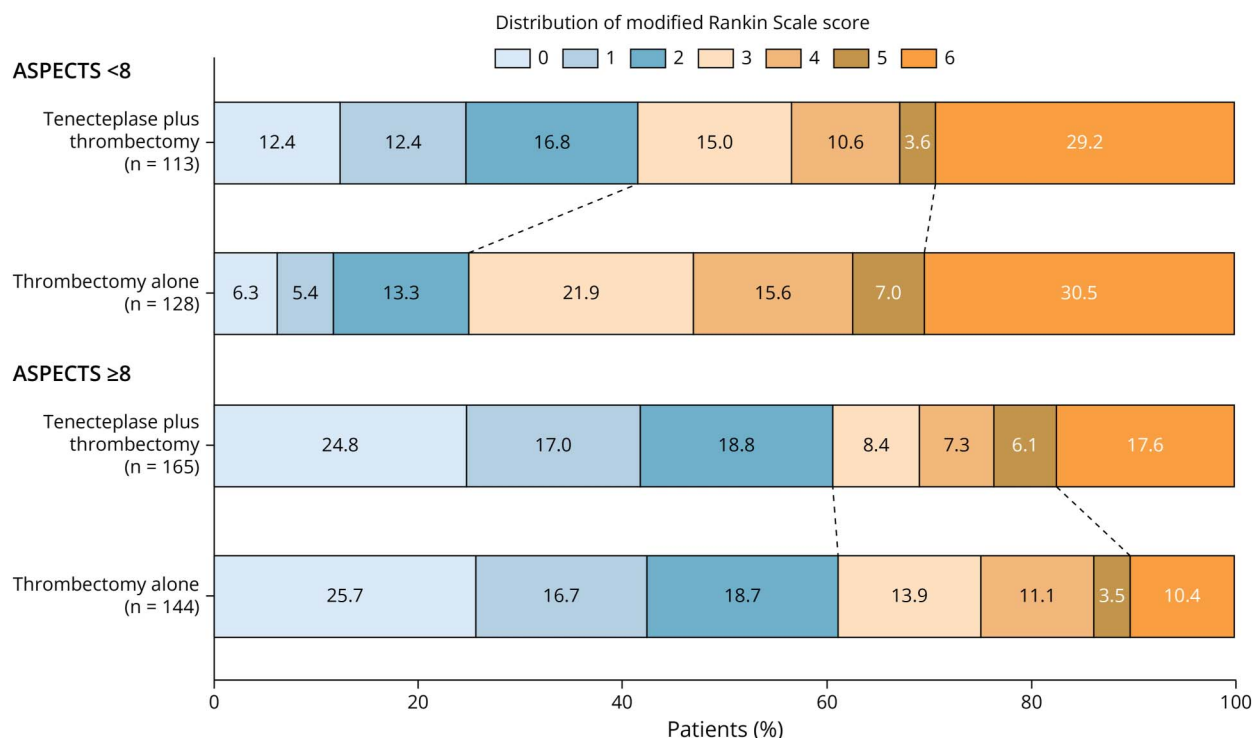
^c Data were not available for 1 patient in the ASPECTS <8 group.

^d Data were not available for 26 patients (14 in the ASPECTS <8 group and 12 in the ASPECTS ≥8 group).

^e Data were not available for 10 patients (6 in the ASPECTS <8 group and 4 in the ASPECTS ≥8 group).

^f Cox proportional hazards regression was used to calculate the hazard ratio with 95% CI for 90-d mortality.

Figure 2 Modified Rankin Scale Score Distribution Stratified by ASPECTS



ASPECTS = Alberta Stroke Program Early Computed Tomography Score.

Discussion

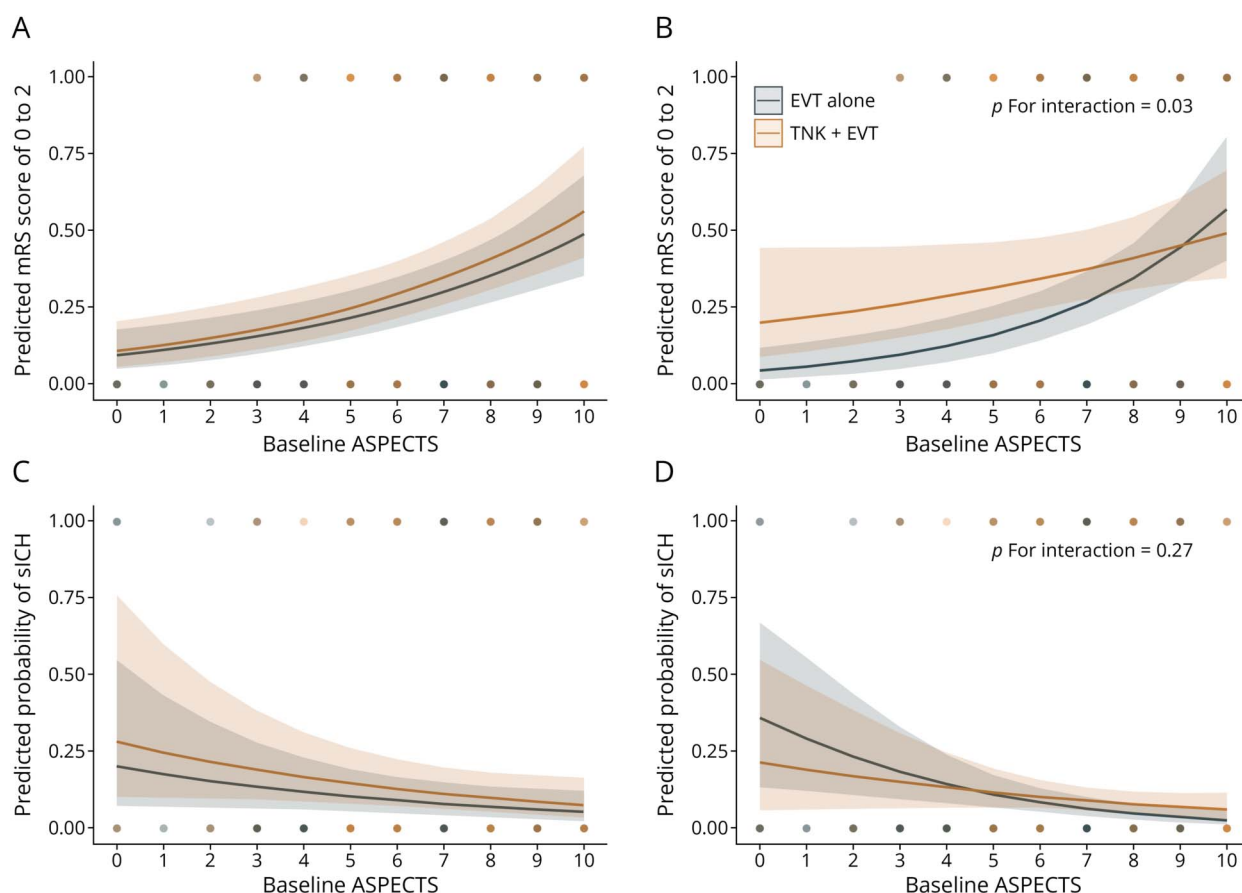
In this study, we performed a post hoc analysis of the BRIDGE-TNK trial to explore the effect modification by baseline ASPECTS on the efficacy and safety of intravenous tenecteplase before thrombectomy in patients with acute large-vessel occlusion stroke presenting within 4.5 hours of symptom onset. Our key finding was a distinct heterogeneous treatment response across ASPECTS subgroups: Among patients with favorable imaging (ASPECTS 8–10), tenecteplase significantly increased prethrombectomy recanalization rates but was associated with numerically higher rates of sICH and mortality, resulting in no net gain in functional independence. By contrast, for patients with ASPECTS 0–7, tenecteplase bridging therapy was associated with a significant improvement in functional independence at 90 days, without a concomitant increase in ICH or all-cause mortality.

Clarifying the impact of baseline ASPECTS on bridging therapy within the 4.5-hour time window is clinically critical because inconsistent imaging criteria across previous trials have hindered the formation of a unified consensus on this strategy. Recent RCTs comparing bridging IVT (predominantly alteplase) with direct thrombectomy have yielded conflicting results: A pooled individual participant data analysis suggested a modest benefit of IVT within 2.3 hours of onset but not beyond this time frame, while many of these

trials either excluded patients with low ASPECTS or enrolled only a small subset, leaving the efficacy and safety of prethrombectomy IVT in patients with unfavorable ASPECTS uncertain.^{1–7} Notably, the BRIDGE-TNK trial did not impose ASPECTS-based eligibility restrictions, enabling robust subgroup analyses by ischemic core extent and filling this evidence gap.

Our findings of ASPECTS-modified treatment effects of prethrombectomy tenecteplase revealed critical discrepancies with prior research by Gabriel Broocks et al.¹⁵ In their retrospective analysis including 720 patients undergoing EVT, higher ASPECTS (6–10) was associated with greater benefit from alteplase bridging therapy. In our analysis of tenecteplase, we observed heterogeneous responses across ASPECTS subgroups. Notably, among patients with favorable ASPECTS (8–10), tenecteplase did not improve 90-day functional outcomes. The reasons for this discrepancy remain unclear. Consistent with prior evidence,^{10–13} tenecteplase may increase early recanalization rates; however, it was combined with a numerically higher incidence of sICH, which possibly offset its net benefit. It is of importance to note that functional independence was achieved in 60% of patients in both treatment groups. We hypothesize that this paradox may also be explained by the inherent clinical characteristics of patients with small ischemic cores, who often present early (median 79 minutes from onset) and have robust collaterals (28.8% with good collaterals), making near-maximal penumbral

Figure 3 Effect of IV Tenecteplase Before Thrombectomy According to Baseline ASPECTS



Adjusted predicted probabilities of 90-day mRS of 0–2 (A, B) and 48-hour symptomatic intracranial hemorrhage (sICH; C, D) by baseline ASPECTS and treatment groups. Plots show estimates from regression models without (A, C) and with (B, D) treatment-by-ASPECTS interactions. ASPECTS = Alberta Stroke Program Early Computed Tomography Score; EVT = endovascular therapy; mRS = modified Rankin Scale; TNK = tenecteplase.

salvage possible even without tenecteplase. Thus, for patients with favorable imaging profiles, the net benefits of bridging therapy remain open to interpretation, and clinical decisions should be more cautious and take other factors, such as anticipated delays to EVT, collateral status, and bleeding risk, into consideration.

Conversely, significantly better functional outcomes with tenecteplase bridging therapy were observed in patients with ASPECTS 0–7. In this subgroup, tenecteplase-treated patients had lower rates of rescue interventions and shorter groin puncture-to-reperfusion times compared with those undergoing direct thrombectomy. Shorter procedure time is a well-established predictor of favorable functional outcomes,^{26,27} offering a plausible mechanistic explanation. Patients who have unfavorable ASPECTS in the early time window often harbor more tenuous collaterals, as observed in our study and prior literature.²⁸ Such fast progressors may be prone to continued rapid infarct expansion, making accelerated reperfusion to terminate the ischemic cascade particularly impactful.^{29,30} This is consistent with the hypothesis of Lin et al.³¹ that patients with rapid core growth

are more likely to benefit from bridging IVT, an effect potentially related to preprocedural IVT facilitating clot lysis and shortening puncture-to-reperfusion time. Of note, this procedural advantage was achieved without delaying groin puncture, addressing a key concern of bridging therapy. Collectively, these findings suggested that IV tenecteplase before thrombectomy might be reasonable for patients with unfavorable imaging when treated within 4.5 hours of onset.

Regarding safety, sICH remains the primary concern for bridging therapy in patients with unfavorable ASPECTS. Our data showed no significant difference in sICH or any ICH rates between 2 treatment groups in patients with ASPECTS 0–7, with consistent results observed after propensity score matching and sensitivity analysis restricted to anterior circulation large-vessel occlusions using only NCCT-based ASPECTS. The sICH rate was 10.0% in tenecteplase-treated patients with ASPECTS 0–7 and 15.6% in those with ASPECTS 0–5, which is comparable with the 13.4% reported in a prior observational study of tenecteplase bridging therapy in large core infarcts (ASPECTS 0–5).³² A

Table 3 Characteristics and Outcomes After Propensity Score Matching

Variables, no. (%)	ASPECTS <8				ASPECTS ≥8			
	Tenecteplase plus thrombectomy (n = 90)	Thrombectomy alone (n = 90)	p Value	SMD	Tenecteplase plus thrombectomy (n = 110)	Thrombectomy alone (n = 110)	p Value	SMD
Age, median (IQR), y	69 (60–77)	69 (64–75)	0.76	0.055	71 (58–76)	70 (63–76)	0.58	0.100
Sex, male	49 (54.4)	49 (54.4)	1.00	<0.001	63 (57.3)	64 (58.2)	0.89	0.018
Medical history								
Hypertension	60 (66.7)	61 (67.8)	0.87	0.024	71 (64.5)	67 (60.9)	0.58	0.075
Diabetes	23 (25.6)	21 (23.3)	0.73	0.052	18 (16.4)	21 (19.1)	0.60	0.071
Hyperlipidemia	20 (22.2)	20 (22.2)	1.00	<0.001	24 (21.8)	23 (20.9)	0.87	0.022
Atrial fibrillation	37 (41.1)	33 (36.7)	0.54	0.091	31 (28.2)	33 (30.0)	0.77	0.040
Ischemic stroke	20 (22.2)	18 (20.0)	0.72	0.054	14 (12.7)	14 (12.7)	1.00	<0.001
Coronary heart disease	21 (23.3)	20 (22.2)	0.86	0.026	18 (16.4)	21 (19.1)	0.60	0.071
Prestroke mRS >1	0	0	NA	<0.001	0	0	NA	<0.001
Baseline systolic blood pressure, median (IQR), mm Hg	147 (133–161)	145 (131–165)	0.92	0.007	143 (129–166)	146 (123–167)	0.88	0.055
Baseline glucose, median (IQR), mg/L	7.5 (6.0–9.4)	7.5 (6.5–9.0)	0.79	0.060	7.0 (5.9–8.4)	7.1 (6.2–8.5)	0.49	0.084
Baseline NIHSS score, median (IQR)	18 (12–20)	18 (13–21)	0.74	0.047	15 (12–19)	14 (11–20)	0.44	0.074
ASPECTS assessment mode			1.00	<0.001			1.00	<0.001
CT	77 (85.6)	77 (85.6)			100 (90.9)	100 (90.9)		
MRI	13 (14.4)	13 (14.4)			10 (9.1)	10 (9.1)		
Ischemic location			0.89	0.071			1.00	<0.001
Left hemisphere	38 (42.2)	41 (45.6)			54 (49.1)	54 (49.1)		
Right hemisphere	44 (48.9)	42 (46.7)			49 (44.5)	49 (44.5)		
Posterior circulation	8 (8.9)	7 (7.8)			7 (6.4)	7 (6.4)		
Occlusion location			0.77	0.158			0.94	0.086
Internal carotid artery	35 (38.9)	38 (42.2)			26 (23.6)	30 (27.3)		
Middle cerebral artery M1	42 (46.7)	42 (46.7)			68 (61.8)	64 (58.2)		
Middle cerebral artery M2	6 (6.7)	3 (3.3)			9 (8.2)	9 (8.2)		
Vertebrobasilar artery	7 (7.8)	7 (7.8)			7 (6.4)	7 (6.4)		
Stroke etiology			0.89	0.073			0.99	0.020
Large artery atherosclerosis	27 (30.0)	27 (30.0)			35 (31.8)	36 (32.7)		
Cardioembolism	52 (57.8)	54 (60.0)			66 (60.0)	65 (59.1)		
Undetermined or others	11 (12.2)	9 (10.0)			9 (8.2)	9 (8.2)		
From onset to hospital	183 (133–226)	178 (136–221)	0.82	0.030	152 (123–211)	157 (126–199)	0.81	0.030
From onset to randomization	122 (74–157)	119 (81–150)	0.89	0.008	81 (49–120)	74 (49–122)	0.80	0.003
From door to groin puncture	76 (61–102)	83 (67–103)	0.56	0.026	89 (72–113)	84 (70–111)	0.42	0.071
From groin puncture to reperfusion	56 (35–84)	65 (45–99)	0.11	0.196	54 (36–87)	65 (40–95)	0.11	0.065
ASITN/SIR score			1.00	<0.001			1.00	<0.001

Continued

Table 3 Characteristics and Outcomes After Propensity Score Matching (*continued*)

Variables, no. (%)	ASPECTS <8				ASPECTS ≥8			
	Tenecteplase plus thrombectomy (n = 90)	Thrombectomy alone (n = 90)	p Value	SMD	Tenecteplase plus thrombectomy (n = 110)	Thrombectomy alone (n = 110)	p Value	SMD
Poor 0-2	73 (81.1)	73 (81.1)			81 (73.6)	81 (73.6)		
Good 3-4	17 (18.9)	17 (18.9)			29 (26.4)	29 (26.4)		
Total attempts, median (IQR)	2 (1-2)	2 (1-3)	0.41	0.185	1 (1-2)	1 (1-2)	0.82	0.025
Rescue therapy	13 (15.3)	23 (26.7)	0.07	0.284	26 (24.5)	29 (26.9)	0.70	0.053
IV tirofiban after thrombectomy	18 (20.0)	30 (33.3)	0.04	0.305	36 (32.7)	36 (32.7)	1.00	<0.001
Outcomes, no. (%)								
mRS score of 0-2 at 90 d	34 (37.8)	18 (20.0)	0.009	0.400	70 (63.6)	66 (60.0)	0.58	0.075
mRS score at 90 d, median (IQR)	3 (2-6)	4 (3-6)	0.09	0.259	2 (1-4)	2 (0-3)	0.78	0.090
mRS score of 0-1 at 90 d	20 (22.2)	8 (8.9)	0.01	0.374	51 (46.4)	47 (42.7)	0.59	0.073
mRS score of 0-3 at 90 d	50 (55.6)	40 (44.4)	0.14	0.224	79 (71.8)	84 (76.4)	0.44	0.104
Successful reperfusion before thrombectomy	5 (5.6)	1 (1.1)	0.10	0.250	7 (6.4)	1 (0.9)	0.03	0.295
Successful reperfusion at final angiography	80 (88.9)	85 (94.4)	0.18	0.202	104 (94.5)	103 (93.6)	0.78	0.039
Any intracranial hemorrhage within 48 h	38 (43.7)	37 (42.0)	0.83	0.033	25 (23.4)	21 (19.1)	0.44	0.105
Symptomatic intracranial hemorrhage within 48 h	10 (11.5)	10 (11.4)	0.98	0.004	9 (8.4)	3 (2.7)	0.07	0.250
Mortality within 90 d	27 (30.0)	26 (28.9)	0.87	0.024	19 (17.3)	10 (9.1)	0.07	0.244

Abbreviations: ASPECTS = Alberta Stroke Program Early Compute Tomography Score; ASITN/SIR grade = the American Society of Interventional and Therapeutic Neuroradiology/Society of Interventional Radiology collateral score; IQR = interquartile range; mRS = modified Rankin Scale; NIHSS = National Institutes of Health Stroke Scale; SMD = standardized mean difference.

post hoc analysis of the ANGEL-ASPECT trial further demonstrated that, in patients with large-core infarction treated with endovascular thrombectomy, the rate of parenchymal hemorrhage was comparable between the bridging IVT plus thrombectomy group and the direct thrombectomy group (14.1% vs 17.0%).³³ Of note, the post hoc analysis of the LASTE trials identified a slight numerical rise in sICH in patients treated with bridging IVT followed by thrombectomy (13.2% vs 7.7%); however, such disparity was limited by the relatively small sample size and did not reach statistical significance.³⁴ Collectively, these findings do not suggest a prohibitive safety signal for intravenous thrombolysis administered before endovascular thrombectomy. Nevertheless, the potential higher risk of sICH observed in this high-risk subgroup still warrants close clinical attention.

Extensive early CT hypodensity indicates elevated water content secondary to ionic edema rather than irreversible

neuronal necrosis.³⁵ Delayed blood-brain barrier disruption is a key driver of hemorrhagic transformation,³⁶ a sequential event in this pathologic cascade. Prompt reperfusion therapy can therefore not only salvage viable cerebral tissue³⁷ but also prevent blood-brain barrier disruption,³⁶ thereby attenuating the established association between extensive early CT hypodensity and hemorrhagic transformation.

Numerous studies have defined an ASPECTS ≤5 as a hallmark of large-core infarction.³⁸⁻⁴³ However, in the present study, we found that certain patients with extensive hypodense lesions on baseline CT or MRI of ASPECTS 3-5 within 4.5 hours of symptom onset could still attain favorable functional outcomes after timely reperfusion therapy (eFigure 2). Of note, no patients with ASPECTS 0-2 attained functional independence. These observations may suggest that ASPECTS 3-5 does not invariably denote an established large-core infarction or confirm its inevitable development;

instead, it signifies a high risk of progression to a large-core infarction in the absence of prompt reperfusion intervention. Notably, this subset of patients may derive meaningful clinical benefit from timely IV tenecteplase combined with EVT.

Historically, large infarction cores were considered a contraindication to both EVT and IVT.⁴⁴ However, recent landmark trials have definitively established the efficacy of EVT in patients with large infarcts (predominantly ASPECTS 3–5), marking a paradigm shift in the management of this high-risk subgroup.^{38–43} Aligning with these advances, we investigated the efficacy of tenecteplase bridging therapy in early-presented patients with low ASPECTS (0–5), 85% of whom had ASPECTS 3–5. In this cohort, tenecteplase bridging yielded a numerically higher rate of functional independence, contrasting with the suggestion by Gabriel Brooks et al. that an ASPECTS of 6 may represent a threshold for alteplase benefit,¹⁵ as well as with a meta-analysis of 6 RCTs indicating no alteplase benefit in patients with ASPECTS 0–5.⁸ However, our results align with a retrospective study of patients with DWI-ASPECTS 0–5, in which tenecteplase bridging yielded superior 3-month outcomes and lower mortality than alteplase.³² Notably, the rates of functional independence (27.3%) and mortality (30.3%) in our tenecteplase cohort were almost identical to those in that study (28.4% and 26.8%, respectively). Nevertheless, these results should be interpreted with caution given the exploratory nature of this subgroup analysis and limited sample size.

Several limitations should be acknowledged. First, this study is a post hoc subgroup analysis with multiple comparisons unadjusted for multiplicity. The findings are therefore exploratory and hypothesis generating. Second, our cohort exhibits inherent imaging heterogeneity across NCCT, DWI, and pc-ASPECTS modalities. Although this heterogeneity mirrors real-world clinical practice, discrepant sensitivity in infarct volume quantification may introduce analytical bias. To address this limitation, we performed a sensitivity analysis confined exclusively to patients with anterior circulation occlusion evaluated solely by NCCT, which validated our primary findings. Third, despite blinded review by a core imaging laboratory, the ASPECTS assessment within 4.5 hours of stroke onset remains challenging owing to early time points, variable image quality, and interobserver variability. CT perfusion imaging could provide a more accurate evaluation of ischemic core, but it was not routinely performed within the 4.5-hour time window in our clinical setting, with only 144 patients available. Moreover, quantitative NCCT hypodensity volumetry, as applied in the SELECT-2 trial for patients with large ischemic cores and extended time windows,⁴⁵ was challenging in our cohort given the indistinct ischemic boundaries on hyperacute imaging. These factors limit our ability to evaluate the influence of ischemic core volume on the efficacy of prethrombectomy tenecteplase therapy. Fourth, we did not investigate whether detailed hypoattenuation grading on NCCT (early vs clear hypoattenuation) can guide tenecteplase administration. Future studies

incorporating net water uptake analysis may further optimize patient selection in individuals with low ASPECTS.⁴⁶ Fifth, clot length (a potential modifier of IVT response) was not measured. Nonetheless, clot occlusion location and etiology were balanced between groups. Finally, all patients were Chinese, limiting generalizability to other ethnic populations.

In this subgroup analysis of the BRIDGE-TNK trial, intravenous tenecteplase administered before thrombectomy might be effective in improving functional outcomes among patients with an ASPECTS <8 presenting within 4.5 hours of symptom onset, without an obvious increase in sICH. For patients with an ASPECTS of 8–10, although intravenous tenecteplase was associated with a higher rate of early recanalization, the numerically higher risk of bleeding and mortality warrants careful clinical consideration. Further validation through randomized controlled trials is needed.

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Author Contributions

X. Huang: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. J. Xu: major role in the acquisition of data. J.L. Saver: drafting/revision of the manuscript for content, including medical writing for content. H. Xu: major role in the acquisition of data. G. Yuan: major role in the acquisition of data. Z. Kong: major role in the acquisition of data. Z. Ruan: major role in the acquisition of data. C. Li: major role in the acquisition of data. G. Yang: major role in the acquisition of data. G.W. Albers: study concept or design. T.N. Nguyen: drafting/revision of the manuscript for content, including medical writing for content. H. Sang: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. Z. Qiu: major role in the acquisition of data; study concept or design. Q. Yang: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. Z. Zhou: drafting/revision of the manuscript for content, including medical writing for content. Z. Li: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data.

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Disclosure

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